

An exact sparse reduction from boxed linear feasibility to a packing threshold

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Abstract

Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $U > 0$. We give a deterministic sparse reduction from deciding whether

$$Ax \leq b, \quad x \in [-U, U]^n,$$

is feasible to testing whether the optimum of a packing linear program with entrywise nonnegative data reaches a specified threshold. The output has at most $2n$ variables, its constraint matrix P satisfies $\text{nnz}(P) \leq \text{nnz}(A) + 2n$, and it is constructed in $O(\text{nnz}(A) + m + n)$ exact sparse-arithmetic operations and output writes. An explicit packing point at the threshold decodes to a boxed feasible point in linear time.

The prompting motivation was the quantitative reduction of Kyng, Wang, and Zhang from accurate packing-LP solving to linear-system solving. Their source problem, hypotheses, and approximate guarantee are different; the theorem here is neither a specialization nor a strengthening of theirs. The constructions share only broad algebraic motifs: split signed variables, nonnegative shifts, and an objective that forces bounding inequalities to be tight. Our exact construction creates no uniform objective gap: infeasible boxed systems can map to packing optima arbitrarily close to the threshold. Thus it does not by itself transfer running times for multiplicative packing-LP approximation.

Note. This paper was fully AI generated with ChatGPT 5.5 Pro from prompts by Xinyuan Cao, Hanxi Lin, and Richard Peng. The present repository version, including this disclosure note, was audited and revised by GPT-5 in the Codex harness.

1 Introduction

For $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and $U > 0$, consider the boxed linear-feasibility problem

$$Ax \leq b, \quad -U\mathbf{1} \leq x \leq U\mathbf{1}. \tag{1}$$

The entries of A may have either sign. The target of this paper is a packing linear program

$$\max\{e^\top z : z \geq 0, Pz \leq d\}, \tag{2}$$

in which P , d , and e are entrywise nonnegative. Our goal is not to solve either program, but to convert (1) to an exact objective-threshold question for (2) while preserving sparsity.

For rational input, general linear programming is polynomial-time solvable by the ellipsoid and interior-point frameworks [Kha79, Kar84]. That bit-complexity statement is distinct

from the unit-cost construction bound proved here. Exact positive linear programming also has a parallel-complexity barrier: Trevisan and Xhafa proved that exact positive-LP solution is P-complete, so an NC algorithm would imply $P = NC$ [TX98].

Much of the algorithmic literature therefore concerns approximation. For a packing program with optimum OPT, a $(1 - \varepsilon)$ -approximate solution is a feasible point of value at least $(1 - \varepsilon) \text{OPT}$. Luby and Nisan initiated width-independent parallel approximation for positive linear programs [LN93]; Plotkin, Shmoys, and Tardos developed the Lagrangian-relaxation framework for fractional packing and covering [PST95]. For an explicit constraint matrix with N nonzeros, r rows, and c columns, Koufogiannakis and Young gave, with high probability, feasible primal and dual solutions within a factor $1 + \varepsilon$ of optimum in

$$O\left(N + \frac{(r + c) \log N}{\varepsilon^2}\right)$$

time [KY14]. Allen-Zhu and Orecchia subsequently obtained $\tilde{O}(N/\varepsilon)$ time for packing and $\tilde{O}(N/\varepsilon^{3/2})$ time for covering [AZO19].

The prompt that led to this note referred to the work of Kyng, Wang, and Zhang on the accuracy dependence of packing-LP solvers [KWZ20]. They reduce conditioned least-squares problems to packing programs and show that sufficiently fast, sufficiently accurate packing solvers would improve algorithms for linear systems. Their dense and sparsity-preserving theorems assume full column rank and a known upper bound on the spectral condition number. That bound supplies a solution scale; a condition-dependent choice of packing accuracy then permits residual decoding and iterative refinement.

The result below addresses a different, elementary question. Its input is an arbitrary signed inequality system together with an explicit coordinate bound U , and its target is an exact packing threshold. It is not extracted as a special case of the reduction of Kyng, Wang, and Zhang, nor does either theorem imply the other. The limited connection is that both constructions use split variables, bounding constraints, and objective tightness to remove signs. We make no novelty claim for those general devices. The contribution here is a self-contained exact formulation, sharp sparse accounting, and an explicit statement of what near-threshold points do and do not imply.

Our main result is the following.

Theorem 1.1 (Exact sparse boxed-feasibility reduction). *Let m, n be positive integers, let $A \in \mathbb{R}^{m \times n}$ be given in sparse row form, let $b \in \mathbb{R}^m$, and let $U > 0$. In the exact sparse arithmetic model, a deterministic algorithm constructs, using $O(\text{nnz}(A) + m + n)$ arithmetic operations and output writes, a packing linear program*

$$\max\{e^\top z : z \in \mathbb{R}_{\geq 0}^q, Pz \leq d\}$$

and a threshold $\theta > 0$ such that P , d , and e are entrywise nonnegative,

$$q \leq 2n, \quad \text{nnz}(P) \leq \text{nnz}(A) + 2n,$$

and

$$\exists x \in [-U, U]^n \text{ with } Ax \leq b \iff \max\{e^\top z : z \geq 0, Pz \leq d\} \geq \theta.$$

In the nontrivial branch the output has $q = 2n$, exactly $m + n$ constraints, $\theta = nU$, and $\text{nnz}(P) = \text{nnz}(A) + 2n$. A boxed feasible point can be encoded as a packing point of value θ in $O(n)$ operations; conversely, any explicit feasible packing point of value at least θ decodes to a boxed feasible point in $O(n)$ operations.

Proof idea. Write $x_j = y_j - s_j$ with $y_j, s_j \geq 0$ and intend to enforce $y_j + s_j = U$. For row i , put

$$R_i = \sum_j |A_{ij}|, \quad p_{ij}^+ = A_{ij} + |A_{ij}|, \quad p_{ij}^- = -A_{ij} + |A_{ij}|.$$

The shifted row

$$\sum_j p_{ij}^+ y_j + \sum_j p_{ij}^- s_j \leq b_i + U R_i$$

has nonnegative coefficients and, on the intended slice, is exactly $A_i x \leq b_i$. The packing constraints $y_j + s_j \leq U$ and the objective $\sum_j (y_j + s_j)$ with threshold nU force every one of these inequalities to be tight. If some shifted right-hand side is negative, the corresponding original row is infeasible throughout the box and the algorithm returns a fixed rejecting instance.

Scope. The result concerns construction cost in an exact unit-cost arithmetic model; it gives no bit-complexity or numerical-stability guarantee and does not include the cost of solving the output LP. It is also an exact-threshold reduction, not a promise-gap reduction. We prove in [Section 4](#) that near-threshold packing points decode to additively near-feasible boxed points, but also that infeasible instances can have packing optimum arbitrarily close to nU . Thus a value-only threshold oracle does not supply a witness, and a standard multiplicative approximation does not decide arbitrary boxed feasibility.

The remainder of the paper is organized as follows. [Section 2](#) records the algebraic identity behind the shift. [Section 3](#) gives the algorithm and proves [Theorem 1.1](#). [Section 4](#) gives the approximate decoding bound and the no-gap example.

2 Setup

All vector inequalities are coordinatewise. For a positive integer k , write $[k] = \{1, \dots, k\}$, and let $\mathbf{1}$ denote an all-ones vector of the dimension determined by context. Empty sums are zero.

Definition 2.1 (Packing threshold instance). *A packing linear program has the form*

$$\max\{e^\top z : z \in \mathbb{R}_{\geq 0}^q, Pz \leq d\},$$

where P , d , and e are entrywise nonnegative. A packing threshold instance also specifies $\theta > 0$ and accepts if the maximum is at least θ .

Sparse arithmetic model. The input matrix is available in sparse row form. Reading and writing a stored coefficient, adding or comparing real numbers, taking an absolute value, and multiplying by U each cost one operation. These are exact operations. The cost of writing the sparse output and its dense right-hand side and objective vectors is included.

For $i \in [m]$ and $j \in [n]$, define

$$R_i = \sum_{j=1}^n |A_{ij}|, \quad p_{ij}^+ = A_{ij} + |A_{ij}|, \quad p_{ij}^- = -A_{ij} + |A_{ij}|, \quad c_i = b_i + UR_i. \quad (3)$$

The quantities R_i, p_{ij}^+, p_{ij}^- are nonnegative. The right-hand side c_i need not be.

Observation 2.2 (Shift identity and tight slice). *For all $y, s \in \mathbb{R}_{\geq 0}^n$ and $i \in [m]$,*

$$\sum_{j=1}^n p_{ij}^+ y_j + \sum_{j=1}^n p_{ij}^- s_j = \sum_{j=1}^n A_{ij}(y_j - s_j) + \sum_{j=1}^n |A_{ij}|(y_j + s_j). \quad (4)$$

If $y_j + s_j \leq U$ for every j and $\sum_j (y_j + s_j) \geq nU$, then $y_j + s_j = U$ for every j .

Proof. Expanding the definitions of p_{ij}^+ and p_{ij}^- gives (4). For the second assertion, set $\delta_j = U - y_j - s_j \geq 0$. The objective inequality says $\sum_j \delta_j \leq 0$, so every nonnegative δ_j is zero. $\square$

The tight slice is in bijection with the box. If $x \in [-U, U]^n$, then

$$y_j = \frac{U + x_j}{2}, \quad s_j = \frac{U - x_j}{2} \quad (5)$$

is nonnegative, $y_j + s_j = U$, and $y_j - s_j = x_j$. Conversely, nonnegative y_j, s_j with sum U give $x_j = y_j - s_j \in [-U, U]$.

3 The reduction

We first describe the algorithm. Coefficients equal to zero are not stored.

Algorithm BoxToPACKING. Given A, b, U , perform the following steps.

1. Compute R_i and c_i from (3) for every $i \in [m]$.
2. If $c_i < 0$ for some i , return the one-variable packing program

$$\max\{0 \cdot z : z \geq 0, z \leq 0\}$$

with threshold $\theta = 1$.

3. Otherwise, use variables

$$z = (y_1, \dots, y_n, s_1, \dots, s_n) \in \mathbb{R}_{\geq 0}^{2n},$$

set the objective vector to the all-ones vector, and set $\theta = nU$.

4. For each $i \in [m]$, create the shifted row

$$\sum_{j=1}^n p_{ij}^+ y_j + \sum_{j=1}^n p_{ij}^- s_j \leq c_i. \quad (6)$$

5. For every $j \in [n]$, create the box row

$$y_j + s_j \leq U. \quad (7)$$

Return these $m + n$ constraints, the objective, and the threshold.

The exceptional branch is justified by the following elementary certificate.

Lemma 3.1 (A negative shifted right-hand side certifies infeasibility). *If $c_i < 0$ for some $i \in [m]$, then the boxed system is infeasible.*

Proof. For every $x \in [-U, U]^n$,

$$(Ax)_i \geq -\sum_{j=1}^n |A_{ij}| |x_j| \geq -UR_i.$$

The inequality $c_i = b_i + UR_i < 0$ is equivalent to $b_i < -UR_i$. Hence $(Ax)_i > b_i$ throughout the box. $\square$

We next prove the exact equivalence in the main branch.

Lemma 3.2 (Threshold equivalence and witness conversion). *Assume $c_i \geq 0$ for every $i \in [m]$. The boxed system is feasible if and only if the packing program defined by (6) and (7) has optimum at least nU . The maps $x \mapsto (y, s)$ from (5) and $(y, s) \mapsto y - s$ convert explicit witnesses in $O(n)$ operations.*

Proof. Suppose $x \in [-U, U]^n$ and $Ax \leq b$. Define y, s by (5). They satisfy every box row with equality. By (4), for each $i \in [m]$,

$$\sum_j p_{ij}^+ y_j + \sum_j p_{ij}^- s_j = (Ax)_i + UR_i \leq b_i + UR_i = c_i.$$

Thus (y, s) is packing-feasible and has objective nU .

Conversely, the origin is packing-feasible because every c_i is nonnegative. The box rows make the feasible region compact, so the maximum is attained. Let (y, s) be feasible with objective at least nU . The box rows also bound that objective by nU ; hence Observation 2.2 gives $y_j + s_j = U$ for every j . Put $x = y - s$. Then $x \in [-U, U]^n$, and (4) gives

$$(Ax)_i + UR_i = \sum_j p_{ij}^+ y_j + \sum_j p_{ij}^- s_j \leq c_i = b_i + UR_i.$$

Therefore $Ax \leq b$. Both conversions require constant work per coordinate. $\square$

Proof of Theorem 1.1. Run BOXTOPACKING. If the exceptional branch is taken, its packing optimum is $0 < 1 = \theta$, while Lemma 3.1 says that the boxed system is infeasible. The branch has one variable, one constraint, and one matrix nonzero; because $n \geq 1$,

$$1 \leq \text{nnz}(A) + 2n.$$

In the main branch, every coefficient in (6) is nonnegative, every c_i is nonnegative, and the box rows, objective, and threshold are positive or nonnegative as required. Thus the output is a packing threshold instance. Lemma 3.2 proves the exact equivalence and the witness statements.

It remains to count the output and the construction work. If $A_{ij} > 0$, then $p_{ij}^+ = 2A_{ij}$ and $p_{ij}^- = 0$; if $A_{ij} < 0$, then $p_{ij}^+ = 0$ and $p_{ij}^- = -2A_{ij}$. Consequently, each stored nonzero of A produces exactly one nonzero in the shifted rows. The n box rows contribute two nonzeros each, so

$$\text{nnz}(P) = \text{nnz}(A) + 2n$$

in the main branch. A constant number of sparse scans computes the row radii and writes the shifted coefficients. Initializing the m right-hand sides, the $2n$ objective entries, and the n box rows costs $O(m+n)$ more operations and writes. This proves the claimed $O(\text{nnz}(A) + m + n)$ bound. $\square$

4 The approximation boundary

Although the exact reduction has no promise gap, its feasible points have a simple quantitative interpretation below the threshold.

Proposition 4.1 (Decoding a near-threshold point). *Consider a main-branch instance and any packing-feasible point (y, s) . Define*

$$x = y - s, \quad \delta_j = U - y_j - s_j, \quad \Delta = \sum_{j=1}^n \delta_j = nU - \sum_{j=1}^n (y_j + s_j).$$

Then $x \in [-U, U]^n$, and for every $i \in [m]$,

$$(Ax)_i \leq b_i + \sum_{j=1}^n |A_{ij}| \delta_j \leq b_i + \|A_{i,*}\|_\infty \Delta. \quad (8)$$

In particular, objective value at least $(1 - \varepsilon)nU$ implies

$$(Ax)_i \leq b_i + \varepsilon nU \|A_{i,*}\|_\infty \quad (i \in [m]).$$

Proof. Nonnegativity and the box rows give $|x_j| = |y_j - s_j| \leq y_j + s_j \leq U$. Rearranging the shifted-row inequality with (4) yields

$$(Ax)_i \leq b_i + UR_i - \sum_j |A_{ij}|(y_j + s_j) = b_i + \sum_j |A_{ij}| \delta_j.$$

The second inequality in (8) follows from $\delta_j \geq 0$ and $\sum_j \delta_j = \Delta$. The final statement uses $\Delta \leq \varepsilon nU$. $\square$

The additive bound in Proposition 4.1 does not yield an exact feasibility decision without a separate promise. The next example shows that this is unavoidable for the present construction.

Proposition 4.2 (No uniform objective gap). *For every $\delta \in (0, 1)$, there is an infeasible boxed instance with $m = 2$, $n = 1$, and $U = 1$ whose output takes the main branch and has*

$$\text{OPT} = \theta - \frac{\delta}{2}.$$

Consequently, the separation between infeasible and feasible outputs can be arbitrarily small, both additively and multiplicatively.

Proof. Take

$$A = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ -\delta \end{pmatrix}, \quad U = 1.$$

The boxed system requires $x \leq 0$ and $x \geq \delta$, so it is infeasible. Both row radii are 1, and the shifted right-hand sides are 1 and $1 - \delta$, so the main branch is taken. Its variables are $y, s \geq 0$, its threshold is $\theta = 1$, and its constraints are

$$2y \leq 1, \quad 2s \leq 1 - \delta, \quad y + s \leq 1.$$

The first two bounds are simultaneously attainable and imply

$$\text{OPT} = \frac{1}{2} + \frac{1 - \delta}{2} = 1 - \frac{\delta}{2}.$$

Letting δ tend to zero proves the claim. $\square$

This example also marks the limit of the comparison with Kyng, Wang, and Zhang. Their linear-system consequence uses additional condition-dependent scaling, residual analysis, and iterative refinement. None of those quantitative ingredients follows from the exact equivalence here, and neither result should be read as a version of the other.

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